Medical Lasers

What is a Laser?

Definition:
The word LASER stands for:
Light Amplification by Stimulated Emission of Radiation

This means:
Amplification of light by stimulated emission of radiation.

Physical Principle of Laser:
A laser depends on:
• Exciting atoms
• Forcing them to emit identical photons

This produces:
• Very intense light
• Organized light
• Monochromatic light

Components of a Laser:
1. Active Medium
The material that produces laser light.
It may be:
• Gas
• Liquid
• Crystal
• Semiconductor

Examples:
He-Ne laser → Helium-Neon gas
CO_2 laser → Carbon dioxide gas
Nd:YAG laser → Neodymium crystal
Diode laser → Semiconductor

2. Energy Source (Pump)
Its function is to supply energy to atoms until they become excited.

3. Mirrors
There are two mirrors:
• One fully reflective
• One partially reflective

Their function is amplifying light inside the device.

How Does a Laser Work?
Stage 1 — Energy Absorption
Atoms absorb energy from the source.
The electron moves from a low energy level to a higher energy level.

Stage 2 — Spontaneous Emission
The electron returns automatically and emits a photon.
However, the photons are random.

Stage 3 — Stimulated Emission
If a photon passes near an excited atom, it forces the atom to emit another photon.

The new photon has:
• The same energy
• The same direction
• The same phase
• The same frequency

Result:
A powerful and organized laser beam is produced.

Properties of Laser:
1. Monochromatic
• One color
• One wavelength

2. Coherent
• All waves are in phase

3. Directional
• The beam spreads very little

4. High Intensity
• Very high energy

Comparison Between Ordinary Light and Laser:
Ordinary Light → Random, multiple colors, diverges, low intensity
Laser → Organized, one color, focused, high intensity

Types of Medical Lasers:
CO_2 Laser → Surgery
Nd:YAG Laser → Tumor treatment
Argon Laser → Eye treatment
Excimer Laser → Vision correction
Diode Laser → Skin and dental applications

Interaction with Tissue:
When laser light strikes tissue, the following may occur:
• Reflection
• Absorption
• Scattering
• Transmission

Medical Effects of Lasers:
1. Thermal Effect
Conversion of light into heat.
Used for:
• Cutting tissue
• Cauterization

2. Photochemical Effect
Chemical reaction caused by light.
Used in cancer treatment.

3. Photoablation
Evaporation of tissue.
Used in vision correction surgery.

Medical Applications of Lasers:
1. Ophthalmology (LASIK)
Uses Excimer laser to correct:
• Myopia

• Hyperopia

2. Surgery
Used for:
• Cutting tissues
• Stopping bleeding

Laser reduces bleeding because it seals blood vessels.

3. Tumor Treatment
Uses Nd:YAG laser to destroy tumors.

4. Dermatology
Used for:
• Tattoo removal
• Hair removal
• Pigmentation treatment

5. Dentistry
Used for:
• Gum cleaning
• Removing tooth decay

Advantages of Medical Lasers:
• High precision
• Less bleeding
• Less pain
• Faster healing

Disadvantages of Lasers:
• High cost
• May cause burns
• Eye protection is required

Important Equations:
Photon energy:
$E = hf$

Relationship between wavelength and frequency:
$c = f\lambda$

Medical Example:
If λ = 600 nm

Calculate the frequency:

$f = c / \lambda$

$f = (3 \times 10^8) / (600 \times 10^{-9})$

$f = 5 \times 10^{14}$ Hz

Important Summary:
Monochromatic → One color
Coherent → In phase
Directional → Focused beam
High intensity → High energy

Academic References

1. Webster — Medical Instrumentation

2. Bushberg — Medical Imaging Physics

3. Introduction to Laser Physics — Siegman

4. Medical Laser Applications — Leon Goldman